

THE DFP METHOD

Lecture 20: Presented by Jan van Vuuren



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The DFP update formula (p. 97)

In 1959, William Cooper Davidon (1927–2013) proposed what is today recognised as the earliest quasi-Newton method

His method is now called the DFP method, because it was updated and popularised in 1963 by the British mathematicians Roger Fletcher (1939–2016) and Michael J.D. Powell (1936–2015)

Today the method is, however, less frequently used

Davidon, Fletcher and Powell opted to estimate the inverse $\mathbf{H}^k = (\mathbf{B}^k)^{-1}$ of the Hessian matrix $\nabla^2 f(\mathbf{x}^k)$ instead of attempting to estimate the Hessian matrix

Taking

$$\mathbf{q}^k = \mathbf{s}^k \quad \text{and} \quad \mathbf{r}^k = \mathbf{H}^k \mathbf{y}^k$$

in an update formula of the form

$$\Delta^k = a^k \mathbf{q}^k (\mathbf{q}^k)^T + b^k \mathbf{r}^k (\mathbf{r}^k)^T$$

it can be shown that

$$\mathbf{H}^{k+1} = \mathbf{H}^k + \frac{\mathbf{s}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} - \frac{\mathbf{H}^k \mathbf{y}^k (\mathbf{H}^k \mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{H}^k \mathbf{y}^k}.$$

The DFP update formula (p. 97)

By applying the Sherman-Morrison formula to the update rule

$$\mathbf{H}^{k+1} = \mathbf{H}^k + \frac{\mathbf{s}^k(\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} - \frac{\mathbf{H}^k \mathbf{y}^k (\mathbf{H}^k \mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{H}^k \mathbf{y}^k},$$

it follows that the corresponding update rule involving Hessian matrix approximations is

$$\mathbf{B}^{k+1} = \left(\mathbf{I} - \frac{\mathbf{y}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) \mathbf{B}^k \left(\mathbf{I} - \frac{\mathbf{s}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) + \frac{\mathbf{y}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k}.$$

The DFP method in pseudo-code (p.97)

Algorithm 3.14 (The Davidon-Fletcher-Powell method)

Input: An instance of a multivariate differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ as to an approximate local optimum of f , an initial Hessian estimate $\mathbf{B}^0$ or an initial inverse Hessian estimate $\mathbf{H}^0$, as well as a search termination criterion

Output: An approximation of a local optimum of f close to $\mathbf{x}^0$ satisfying the termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Compute the search direction $\mathbf{p}^k$ by Gauss-Jordan elimination to solve the system of linear equations $\mathbf{B}^k \mathbf{p}^k = -\nabla f(\mathbf{x}^k)$, or compute it directly as $\mathbf{p}^k = -\mathbf{H}^k \nabla f(\mathbf{x}^k)$

Step 4 Compute an appropriate step length α^k in the search direction $\mathbf{p}^k$ by performing an exact line search or a backtracking Armijo-Goldstein line search in respect of the function $f(\mathbf{x}^k + \alpha^k \mathbf{p}^k)$

Step 5 Compute the next search estimate as $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{p}^k$

Step 6 $\mathbf{s}^k \leftarrow \mathbf{x}^{k+1} - \mathbf{x}^k$

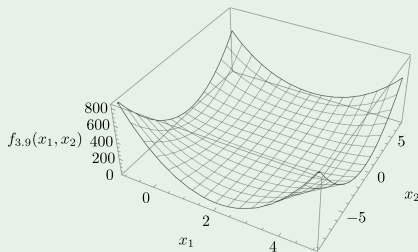
Step 7 $\mathbf{y}^k \leftarrow \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

Step 8 $\mathbf{B}^{k+1} \leftarrow \left(\mathbf{I} - \frac{\mathbf{y}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) \mathbf{B}^k \left(\mathbf{I} - \frac{\mathbf{s}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} \right) + \frac{\mathbf{y}^k (\mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k}$ or
 $\mathbf{H}^{k+1} \leftarrow \mathbf{H}^k + \frac{\mathbf{s}^k (\mathbf{s}^k)^T}{(\mathbf{y}^k)^T \mathbf{s}^k} - \frac{\mathbf{H}^k \mathbf{y}^k (\mathbf{H}^k \mathbf{y}^k)^T}{(\mathbf{y}^k)^T \mathbf{H}^k \mathbf{y}^k}$

Step 9 $k \leftarrow k + 1$ and return to Step 2

Example 3.18 (pp. 98–100)

$$f_{3.9}(x_1, x_2) = 17 - 32x_1 + 24x_1^2 - 8x_1^3 + x_1^4 + 2x_2 + 5x_2^2 - 4x_1x_2^2 + x_1^2x_2^2$$



We apply Algorithm 3.14 (equipped with a Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$) to minimise $f_{3.9}$, starting out from $\mathbf{x}^0 = [-1 \ 1]^T$

We are given an initial symmetric, positive definite Hessian estimate

$$\mathbf{B}^0 = \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$$

We work accurately to six decimal places and terminate the iterative process once the 2-norm $\|\nabla f_{3.9}(\mathbf{x}^k)\|_2$ drops below 0.05 for the first time.

Example 3.18 (pp. 98–100)

Our first step is to compute the gradient vector of $f_{3.9}$ as

$$\nabla f_{3.9}(x_1, x_2) = \begin{bmatrix} 2(x_1 - 2)(8 - 8x_1 + 2x_1^2 + x_2^2) \\ 2 + 2(5 - 4x_1 + x_1^2)x_2 \end{bmatrix}$$

During Iteration $k = 0$, the gradient vector is $\nabla f_{3.9}(\mathbf{x}^0) = [-114 \ 22]^T$

We compute the search direction by solving the system

$$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} p_1^0 \\ p_2^0 \end{bmatrix} = \begin{bmatrix} 114 \\ -22 \end{bmatrix}$$

The first search direction $\mathbf{p}^0 = [\frac{46}{5} \ \frac{37}{10}]^T = [9.2 \ 3.7]^T$ is obtained

A backtracking Armijo-Goldstein line search with parameters $a = 0.4$ and $b = 0.9$ yields the step length $\alpha^0 = 0.205891$

As a result, $\mathbf{s}^0 = \alpha^0 \mathbf{p}^0 \approx [1.894198 \ 0.761797]^T$

Therefore, $\mathbf{x}^1 = \mathbf{x}^0 + \alpha^0 \mathbf{p}^0 \approx [0.894198 \ 1.761797]^T$

Moreover, $\mathbf{y}^0 = \nabla f_{3.9}(\mathbf{x}^1) - \nabla f_{3.9}(\mathbf{x}^0) \approx [101.726656 \ -12.167764]^T$

Example 3.18 (pp. 98–100)

The output of the DFP method coincides with those of Broyden's method, the SR1 method and the BFGS method during the first iteration ...

In anticipation of the update rule

$$B^1 \leftarrow \left(I_2 - \frac{\mathbf{y}^0(\mathbf{s}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} \right) B^0 \left(I_2 - \frac{\mathbf{s}^0(\mathbf{y}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} \right) + \frac{\mathbf{y}^0(\mathbf{y}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0},$$

the following intermediate calculations are first performed during iteration 1:

$$\mathbf{y}^0(\mathbf{s}^0)^T = \begin{bmatrix} 101.726\,656 \\ -12.167\,764 \end{bmatrix} [1.894\,198 \quad 0.761\,797]$$

$$= \begin{bmatrix} 192.690\,428 & 77.495\,061 \\ -23.048\,154 & -9.269\,366 \end{bmatrix},$$

$$\mathbf{s}^0(\mathbf{y}^0)^T = \begin{bmatrix} 1.894\,198 \\ 0.761\,797 \end{bmatrix} [101.726\,656 \quad -12.167\,764]$$

$$= \begin{bmatrix} 192.690\,428 & -23.048\,154 \\ 77.495\,061 & -9.269\,366 \end{bmatrix},$$

$$\mathbf{y}^0(\mathbf{y}^0)^T = \begin{bmatrix} 101.726\,656 \\ -12.167\,764 \end{bmatrix} [101.726\,656 \quad -12.167\,764]$$

$$= \begin{bmatrix} 10\,348.312\,541 & -1\,237.785\,943 \\ -1\,237.785\,943 & 148.054\,481 \end{bmatrix}, \text{ and}$$

$$(\mathbf{y}^0)^T \mathbf{s}^0 = (101.726\,656)(1.894\,198) + (-12.167\,764)(0.761\,797) = 183.421\,062$$

Example 3.18 (pp. 98–100)

The Hessian estimate is consequently updated to

$$\begin{aligned} B^1 &= \left(I_2 - \frac{\mathbf{y}^0(\mathbf{s}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} \right) B^0 \left(I_2 - \frac{\mathbf{s}^0(\mathbf{y}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} \right) + \frac{\mathbf{y}^0(\mathbf{y}^0)^T}{(\mathbf{y}^0)^T \mathbf{s}^0} \\ &= \left(I_2 - \frac{1}{183.421\,062} \begin{bmatrix} 192.690\,428 & 77.495\,061 \\ -23.048\,154 & -9.269\,366 \end{bmatrix} \right) \begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix} \\ &\quad \times \left(I_2 - \frac{1}{183.421\,062} \begin{bmatrix} 192.690\,428 & -23.048\,154 \\ 77.495\,061 & -9.269\,366 \end{bmatrix} \right) \\ &\quad + \frac{1}{183.421\,062} \begin{bmatrix} 10\,348.312\,541 & -1\,237.785\,943 \\ -1\,237.785\,943 & 148.054\,481 \end{bmatrix} \\ &\approx \begin{bmatrix} 56.997\,294 & -8.187\,911 \\ -8.187\,911 & 4.386\,685 \end{bmatrix} \end{aligned}$$

Since

$$\|\nabla f_{3.9}(\mathbf{x}^0)\|_2 \approx 116.103\,402 > 0.05,$$

Algorithm 3.12 does not terminate immediately after the first iteration ...

Example 3.18 (pp. 98–100)

k	$\mathbf{x}^k$	$\nabla f_{3.9}(\mathbf{x}^k)$	$\mathbf{B}^k$	$\mathbf{p}^k$	α^k	$\mathbf{s}^k$	$\mathbf{y}^k$	$\ \nabla f_{3.9}(\mathbf{x}^k)\ _2$
0	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -114 \\ 22 \end{bmatrix}$	$\begin{bmatrix} 14 & -4 \\ -4 & 4 \end{bmatrix}$	$\begin{bmatrix} 9.2 \\ 3.7 \end{bmatrix}$	0.206	$\begin{bmatrix} 1.894 \\ 0.762 \end{bmatrix}$	$\begin{bmatrix} 101.727 \\ -12.168 \end{bmatrix}$	116.103
1	$\begin{bmatrix} 0.894 \\ 1.762 \end{bmatrix}$	$\begin{bmatrix} -12.273 \\ 9.832 \end{bmatrix}$	$\begin{bmatrix} 56.997 & -8.188 \\ -8.188 & 4.387 \end{bmatrix}$	$\begin{bmatrix} -0.146 \\ -2.513 \end{bmatrix}$	1	$\begin{bmatrix} -0.146 \\ -2.513 \end{bmatrix}$	$\begin{bmatrix} 3.018 \\ -11.690 \end{bmatrix}$	15.726
2	$\begin{bmatrix} 0.748 \\ -0.752 \end{bmatrix}$	$\begin{bmatrix} -9.255 \\ -1.858 \end{bmatrix}$	$\begin{bmatrix} 55.001 & -4.390 \\ -4.390 & 4.906 \end{bmatrix}$	$\begin{bmatrix} 0.213 \\ 0.570 \end{bmatrix}$	1	$\begin{bmatrix} 0.214 \\ 0.570 \end{bmatrix}$	$\begin{bmatrix} 4.716 \\ 3.103 \end{bmatrix}$	9.440
3	$\begin{bmatrix} 0.962 \\ -0.181 \end{bmatrix}$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
9	$\begin{bmatrix} 1.962 \\ -1.057 \end{bmatrix}$	$\begin{bmatrix} -0.085 \\ -0.118 \end{bmatrix}$	$\begin{bmatrix} 3.272 & 0.080 \\ 0.080 & 2.011 \end{bmatrix}$	$\begin{bmatrix} 0.025 \\ 0.058 \end{bmatrix}$	1	$\begin{bmatrix} 0.025 \\ 0.058 \end{bmatrix}$	$\begin{bmatrix} 0.058 \\ 0.118 \end{bmatrix}$	0.145
10	$\begin{bmatrix} 1.982 \\ -1.014 \end{bmatrix}$	$\begin{bmatrix} -0.037 \\ -0.029 \end{bmatrix}$	$\begin{bmatrix} 2.061 & 0.042 \\ 0.042 & 6.278 \end{bmatrix}$	$\begin{bmatrix} 0.018 \\ 0.004 \end{bmatrix}$	1	$\begin{bmatrix} 0.018 \\ 0.004 \end{bmatrix}$	$\begin{bmatrix} 0.036 \\ 0.009 \end{bmatrix}$	0.046
11	$\begin{bmatrix} 2.000 \\ -1.009 \end{bmatrix}$	—	—	—	—	—	—	—

Example 3.18 (pp. 98–100)

The DFP method returns $\mathbf{x}^{11} = [1.999\,876 \quad -1.009\,403]^T$ as approximate minimum of $f_{3.9}$; Compare this to the true minimum is $\mathbf{x}^* = [2 \quad -1]^T$

